Dep
UW

Department of Compute Science

Assignment # 03

Course Code: CS-202	**Name:** ______________________________
Course Title: Artificial Intelligence	**Registration No.** ______________________
Total Marks: 20	**Discipline/Semester/Section: BSCS 3rd B**
Assigned Date: 16th January, 2026	**Deadline: 19th January, 2026**

[CLO3, GA4, C4]

Q1 Model the following scenarios using knowledge representation, logical and reasoning to automate the problems:

a. Design an Automated Customer Support Chatbot Using Logic and Reasoning in AI.

Objective:

To understand how logic representation, rule-based reasoning, and inference can be applied in AI by designing a simple chatbot that decides appropriate responses to user queries.

Scenario:

You are building an AI-powered customer support chatbot for an e-commerce website. The chatbot must decide whether to:

- Provide Order Information,
- Provide Return Instructions,
- Connect to a Human Agent,
- Or Do Nothing

Dataset (User Queries & Context):

User Query (Simplified)	Order Status Known?	Query Type	Urgency Level	Action (To Be Inferred)
Q1: "Where is my order?"	Yes	Order Inquiry	Low	?
Q2: "I want to return my shoes."	Yes	Return Request	Medium	?
Q3: "This is urgent, I need help now!"	No	General Help	High	?

Instructions

Step 1: Representation
Represent each user query using simple conditions (e.g., order known, query type, urgency).

Step 2: Logic
Write IF–THEN rules that the chatbot will use to decide actions.

Step 3: Reasoning
Apply the IF–THEN rules to each query.

Step 4: Inference
Fill the "Recommendation" column with the appropriate chatbot action.

Deliverables:

i. Explanation of how user queries are represented using conditions.
ii. IF–THEN rules used by the chatbot.
iii. Step-by-step reasoning for each query.
iv. Final table with inferred recommendations.

b. An AI security assistant monitors a smart home using sensors and rules to detect intrusions or system states.

Knowledge Base:

1. Intrusion Rule:
   If the motion sensor detects movement and the door is open, there is an intruder.
   $\forall x(MotionDetected(x) \land DoorOpen(x) \rightarrow Intruder(x))$

2. Safe Rule:
   If the alarm is not triggered, then there is no intruder.
   $\forall x(\neg Alarm(x) \rightarrow \neg Intruder(x))$

3. Night Light Rule:
   If it is dark, the night light turns on.
   $Dark(x) \rightarrow NightLightOn(x)$

Deliverables:

i. List down the observations.
ii. Can we conclude that an intruder is using Modus Ponens? Why or why not?
iii. What does the system infer about lighting, using deductive reasoning?
iv. What type of reasoning (deductive, inductive, or abductive) would the system need to guess if someone forgot to lock the door, even if no alert is triggered?